1st Lecture 2015

1 The aqueous solution bounded by the plasma membrane of the cell is
(a) cytoplasm (b) nucleolus (c) cytosol (d) cytokine

2 Which of the following is not a cellular organelle ?
(a) ribozyme (b) ribosome (c) golgi bodies (d) lysosome

3 Protein biosynthesis takes place in the
(a) endoplasmic reticulum (b) nucleolus (c) peroxisome (d) ribosome

4 The part of the cell responsible for drug detoxification is the
(a) golgi (b) lysosome (c) endoplasmic reticulum (d) cytosol

5 Macromolecular wastes in the cell are decomposed in the
(a) peroxisome (b) ribosome (c) microsome (d) lysosome

6 Which of the following processes does not take place in the mitochondria?
(a) TCA cycle (b) β-oxidation of fatty acid (c) glycogenesis (d) electron transport chain

ketogenesis is what happens in mitochondria instead.

7 Which of the following conditions is not due to release of lysosomal enzymes ?
(a) postmortem autolysis (b) arthritis (c) inclusion cell disease (d) Scrapie disease

prion infection

8 In cellular protein targeting, the signal molecule carried by enzymes destined for lysosomes is
(a) mannose-6-phosphate (b) glucose-6-phosphate
(c) mannose-3-phosphate (d) ribose-5-phosphate

9 Unwanted free radicals are destroyed in the
(a) peroxisome (b) lysosome (c) golgi bodies (d) mitochondria

10 An organelle that has its own DNA is
(a) endoplasmic reticulum (b) mitochondria it has non-chromosomal DNA
(c) ribosome (d) None of the above

11 Metabolic pathways that partially take place in the mitochondria include all of the followings except
(a) urea cycle (b) ketogenesis (c) gluconeogenesis (d) pyrimidine synthesis

part(1st 2 steps) completely part(1st step) part



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12 Factors affecting fluidity of the plasma membrane do not include
(a) length of fatty acid chain (b) cholesterol content
(c) membrane protein content (d) degree of unsaturation

13 Passive transport of substances across cell membranes does not take place through
(a) ion channels (b) pumps (c) facilitated diffusion (d) simple diffusion

14 In the pong state of the carrier molecule in a facilitated diffusion, the active sites
(a) are exposed to the interior (b) a change in conformation
(c) release the solute into the cell (d) face the exterior

pOng=Outside
pIng=Inside

15 Specialized membrane protein molecules that open like a gate in response to stimulus to allow secretion of substances from the cell are called
(a) pumps (b) ion channels (c) carriers (d) molecular sieve

16 Galactose and mannose are epimers of glucose, they differ from glucose in the configuration around
(a)C1 and C2 respectively (b) Carbon 2and Carbon 4 respectively
(c) Carbon 2 and Carbon3 respectively (d) Carbon 4 and Carbon 2 respectively

17 What is the gllycosidic linkage in Trehalose?
(a) α(1- 4) (b) α(1 - 1) (c) α (1 - 6) (d)α( 1 - 2)

18 What are the monosaccharide constituents of Lactose?
(a) Glucose and Glucose (b) Glucose and Fructose
(c) Galactose and Fructose (d) Galactose and Glucose

19 The D&L isomerism is determined by the configuration on
(a) The anomeric carbon (b) Carbon 5 of all monosaccharide
(c) Cabon 6 of aldoses (d) Penultimate carbon

20 What is the fate of excess carbohydrates in man?
(a) It is converted to fat and stored (b) It is used for energy production immediately
(c) It is converted to amino acids for protein biosynthesis (d) None of the above

De-novo synthesis

21 The occurrence of asymmetric carbon atoms lead to the formation of
(a) Structural Isomerism (b) Optical Isomerism (c) Mutarotation
(d) Stereoisomerism

22. Which of the following disaccharides has no free aldehyde or ketone group?
(a) Maltose (b) Sucrose (c)Lactose (d) No one



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23. Which of the following is the type of glycolytic bond found in cellulose?

(a) β (1 - 4) (b) α(1 - 6) (c) β (1-6) (d) α(1- 4)

24. The specific optical rotation of equilibrium mixture of α and β-D glucose is

(a) 18.7degrees (b) 52.7 degrees (c) 112 degrees (d)25.7degrees

25. Anomerism is produced with reference to

(a) Carbon 1 in aldoses and ketoses (b) carbon 2 in aldose and ketose

(c) carbon1 in ketoses and carbon 2 in aldoses (d) carbon 1 in aldoses and carbon 2 in ketoses

carbonyl carbon

26. When Galactose is reduced, the product is

(a)Mannitol (b) Sorbitol (c) Dulcitol (d) Ribitol

27. When galactose reacts with Nitric acid in the presence of heat, the compound formed is

(a) Mucic acid (b) Mannaric acid (c) Galacturic acid (d) Dulcitol

28. Glucuconic acid is formed when glucose is subjected to

Gluconic acid

(a)Reduction to form sugar alcohol (b) Treated with concentrated sulphuric acid

(c) Oxidation with hypobromous acid (d) Action of dilute alkaline solution

29 In a Laboratory, an unknown fluid was tested with excess phenylhydrazine in acetic acid at boiling temperature; tennis-ball shaped crystals were formed. Which of the following sugars was present in the fluid?

(a) Glucose (b) Lactose (c) Maltose (d) Fructose

30. Which of these polysaccharides is not an homoglycan?

(a) Pectins (b) Starch (c) Cellulose (d) Dextrans

31. The following are derived lipids excepts

(a) Diacyglycerol (b) Free fatty acid (c) Cholesterol (d) Neutral lipid

32. Which of these Lipoprotein is implicated in peripheral vascular disease

(a) HDL (b) VLDL (c) Chylomicron (d) LDL Bad cholesterol

33. The following apoproteins are found in VLDL except

(a) APO – B100 (b) APO – C (c) APO – A (d) APO – E

34. Which of the following is not a phospholipid ?

(a) Cerebroside (b) Cephalin (c) Lecithin (d) Sphingomyelin

Glycolipid



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35. Concerning Geometric stereoisomerism of fatty acids

(a) The trans isomers are commonly found in humans than the cis isomers

(b) The cis isomers constitutes health risk in humans than the trans isomers

(c) The cis-isomers are less common in humans than the trans isomers

(d) The trans isomers constitutes major cardiovascular health risk than the cis isomers

36. All of the following are essential fatty acids except

(a) Linoleic acid (b) Oleic acid (c) Linolenic acid (d) Arachidonic acid

37. Sphingomyelin on hydrolysis yields all the following except

(a) Sphringosine (b) Phosphate (c) Choline (d) Glucose

38. Lipids are transported in the blood as

(a) Cholesterol (b)Fatty acids (c) Lipoproteins (d) Neutral lipids

39. Concerning fatty acids the following is true except

(a) They are the most common components of lipid

(b) The higher the number of double bonds they contain the lower the melting points

(c) Naturally occurring fatty acids occur in odd numbers

(d) They undergo β-hydroxylation to form energy substrates.

40. The amphipatic nature of lipids is exhibited by all of the following except

(a) Sphinomyelin (b) Lecithin (c) Ganglioside (d) Glycerol

41. Which of the following is not a major function of amino acids ?

(a) A major component of neuroendocrine system as neurotransmitters

(b) Metabolic substrate for energy production Last resort, not major

(c) Biosynthesis of urea

(d) Productionnof prophyrins

2. Which of the following naturally occurring amino acid is not an L-Amino acid ?

(a) Histidine (b) Arginine (c) Tryptophan (d) Proline

. The following amino acids have right handed stereoisomerism except D-amino acids

(a) Valine (b) Aspartate (c) Alanine (d) Serine

. The following are examples of branched chain aliphatic amino acids except

(a) Leucine (b) Isoleuvine (c) Phenylanine (d) Valine



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45. Concerning derived amino Acids
(a) They are part of the 20 basic amino acids found in humans
(b) They are all toxic to humans
(c) They are produced as metabolites of steroid hormones
(d) None of the above

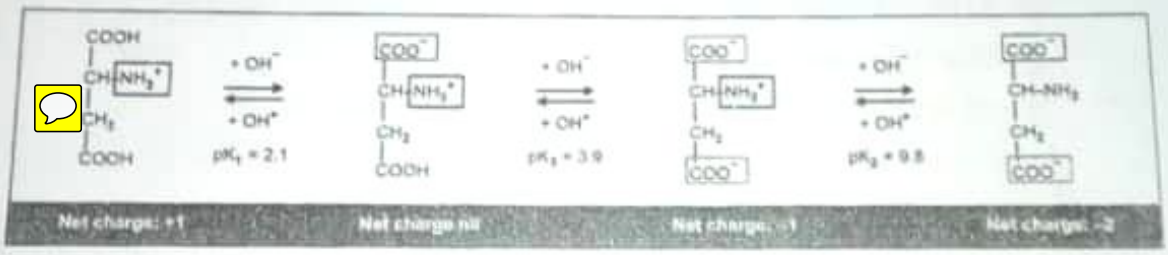
46. Toxic Non-protein amino acids include
(a) Ornithine (b) Di-amino propionic acid
(c) Glutamate-γ-semialdehyde (d) Homocysteine

47. The following amino acids are mainly glucogenic except
(a) Glutamine (b) Valine (c) Lysine (d) Aspartic acid

48. Ketogenic amino acids include the following except
(a) Phenylalanine (b) Leucine (c) Serine (d) Tyrosine

49. The underlisted are not regarded as essential amino acids except
(a) Tryptophan (b) Glutamine (c) Asparagine (d) Arginine

50. Determine the Isoelectric pH (PI) of the amino acid shown below




(a) 6.9 (b) 6.0 (c) 5.3 (d) 3.0

51. Which of the following statement does not describe the characteristics of proteins?
(a) They are synthesized through polymerization of amino acids to form a peptide bonds
(b) They are long polypeptide having more than 50 amino acid residues
(c) Sequencing technique is for quantitative estimation of proteins
(d) Oligopeptides can also function as proteins

2. Concerning Proteins
(a) Incomplete proteins lack all the essential amino acids required for growth and development
lack one
(b) First class proteins can be obtained from both plant and animal sources
only animal
(c) Derived proteins are usually formed from degradation of native proteins
(d) Keratin is a globular protein



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53. Quantitative estimation of protein can be done through the following methods? except??
(a) Spectrophotometer (b) Biuret method
(c) Sanger Method (d) Lowry's method

54. Which of the following is not regarded as a conjugated protein ?
(a) Collagen (b) Haemoglobin (c) Cytochrome (d) Vitellin

55. Which of the following statements is true concerning Quaternary structure of proteins?
(a) Described based on the amino acid sequence
(b) Describes the functional aggregation of polypeptides.
(c) Amino acid residues involved are usually 3-4 residues apart
(d) Describes the 3 dimensional structure of a whole protein

56. Which of the following statement is [redacted] true regarding pH ?
(a) As the pH of a solution increases, the solubility of $CaCO_3$ increase
(b) The solubility of $CaCO_3$ is not affected by the pH of the solution
(c) As the pH of the solution decreases, the solubility of $CaCO_3$ increases
(d) None of the above

57. Which of the following is true about the bicarbonate buffer system?
(a) The ratio of $H_2CO_3$ to $HCO_3^-$ is 20
(b) The ratio of $HCO_3^-$ to $H_2CO_3$ is 20
(c) Increase pH causes the precipitation of $H_2CO_3$
(d) All of the above

58. Which of the following amino acid will most unlikely participate in protein buffer system? ??
(a) Threonine (b) Tyrosine (c) Arginine (d) Methionine

59. Which of the following buffer is suitable for buffering at pH 5?
(A) Phosphate buffer 6.86 (b) Acetate buffer 4.76 (c) Carbonate buffer 3.77 (d) Ammonium buffe 9.25

60. What is the mole ratio between $H_2PO_3$ and $HPO_3^-$ in preparing 0.05 M phosphate Buffer pH 6.5 containing 10 % sucrose ? ( pka = 6.8)
(a) 0.019 : 0.031 (b) 0.0333 : 0.0167 (c) 0.0158 : 0.0342 (d) 0.0269 : 0.0231

61. How many grams of NaOH is required to prepare 500 ml of 0.04M solution?
(a) 0.8 g (b) 0.4 g (c) 0.2 g (d) 0.1 g



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62. How many ml of 0.5 M $H_2SO_4$ are required to make 1500 ml of a 0.002 M $H_2SO_4$ solution?

6ml

?? (a) 0.06 ml (b) 0.4 ml (c) 0.02 ml (d) 1.02 ml

63. What is the $H^+$ ion concentration of 0.01 M $HNO_3$?

(a) 0.0001 mole (b) 0.1 mole (c) 0.01 mole (d) 1 mole

64. The potential hydrogen of the solution above is

(a) 1 (b) 2 (c) 3 (d) 4

65. When a little amount of $H^+$ ion is added to a buffer solution,

(a) The equilibrium shifts to the left (b) The equilibrium shifts to the right

(c)More water molecule is produced (d) None of the above

66. Which of the following is not true about the bicarbonate buffer system?

(a) The kidney helps in maintaining the constant concentration, of $HCO_3^-$ in bicarbonate buffer,

All True??

(b) Carbonic anhydrase helps in removing excess $H_2CO_3$

(c) Carbonic anhydrase helps in producing $H_2CO_3$

(d) Hemoglobin plays a major role in maintaining $H_2CO_3/HCO_3^-$ concentration

67. Which of the following is not true about water?

(a) Its pH changes with increase in temperature ??

(b) It can exist in planar and tetrahedral form

(c)It exhibits hydrophobic interaction with another molecule

(d) It can exhibit dipole-dipole interaction with other molecule

68. Which of the following is a polyprotic acid?

(a) HCl (b) $CH_3COOH$ (c) $HNO_3$ (d) $NaH_2PO_4$

69. Calculate the molar concentration of the conjugate acid in a 50 mM buffer of pH 6.5 (pka of the acid = 6.8)

(a) 0.0167 M (b) 0.0333 M (c) 0.167 M (d) 0.333 M

70. Which of the following acid has many ionization points?

(a) HCl (b) $CH_3COOH$ (c) $HNO_3$ (d) $H_3PO_4$



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71. Which of the following correctly describes the active site of an enzyme?

(a) It is the substrate binding region and contains amino acids from a selected portion of the amino acid sequence
(b) It is a small port lined with amino acid residues that are involved in substrate recognition
(c) It is a hydrophobic region where both substrates and other ligands bind
(d) It is the largest part of the enzyme molecule

72. A coenzyme is a

(a) Protein (b) Carbohydrate (c) Lipid (d) None of these

Select answers for Qs 73-75 from the options given below

(a) Alkylation (b) Carbonyl transfer (c) Carboxylation (d) Oxidation-reduction

What type of reaction is mediated by

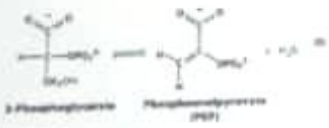
73. Thiamine pyrophosphate? B

74. Glutathione ? D

75. Vitamin B12 ? A

For Questions 76-78, identify the class of enzymes that will catalyze the reactions indicated from the options given below

(a) Oxidoreductase (b) Lyase (c) Ligase (d) Transferase



Enolase, Lyase B

76.

77. Serine dehydratase reaction Lyase

78. Deoxynucleoside triphosphate +$DNA_n$ Deoxynucleoside diphosphate + $DNA_{n+1}$ Transferase

Consider the following model for the enzyme-catalyzed conversion of substrate to product

$$\underbrace{E + S \rightleftharpoons ES}_{\text{Substrate binding}} \xrightarrow{\text{Catalytic step}} E + P$$

Select one of the options below for questions 79-81, where [E] and [S] are the initial concentrations of enzyme and substrate respectively

(a) $ES(K_2+K_3)$ (b) $K_3[ES]$ (c) $K_3[E]$ (d) $K_2+K_3/K_1$



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What is the expression for

79. The rate of breakdown of the ES complexA

80. The maximum steady-state velocity of the reactionC

81. $K_m$ with respect to substrate?D

82. Competitive enzyme inhibition

(a) Occurs when an EI complex is formed which is catalytically more active than the ES complex

(b) Occurs when the binding of the substrate and inhibitor to the enzyme are mutually exclusive T

(c) Produces a pattern of intersecting lines on the 1/v ordinate when data are plotted in double-reciprocal form T

(d) b & c are true

83. The unit of catalytic constant is

(a) Katal (b) μM/Min (c) μM/Min/Mg (d) None of the abmM/min/mol

The options below are for Questions 84 & 85

(a) $K_m/V_{max}$ (b) $K_m$ (c) $1/V_{max}$ (d) None of the above

Which of the options represent

84. The intercept on the X axis for a Woolf plot ?=-Km D

85. The slope of an Eadie-Hofstee plot ? = -Km D

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